Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Gavin Bell	gavinbel@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

My goal is to make a college grade calculator that lets students put in their grades and the weight of the assignment. Then calculate the students current grade in that class.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There are a lot of websites that attempt to address this goal like MarkMyGrades, which also has a grade calculator and it also calculates what grade is needed for remaining work.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I want my program to be easy to use for college students so my implementation will be a simpler version. Instead of trying to make a complete oline grade book, I will focus on entering grades, weights and calculating the current grade. I may also include a target grade feature.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

Calculate a college student's current grade in their course based on their grades and the weights of those grades.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

-Assignment name -Grades -Weight of each assignment -Target grade

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

-Store the grades and weights. -Multiply each grade by its weight. -Add the weighted grades together. -Divide by the total weight that has been entered. -Convert the result into percentage. -If a target grade is entered, calculate the grade needed on the remaining work.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

-Current percentage grade -If a target grade is entered, the grade needed on the remaining work

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

-Students
-Course
-Assignment
-Grade
-Weight
-Target grade
-Due date

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details I need: Assignment, grade, weight, target weight
Details I will ignore: Due date, course

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The most difficult part is going to be making sure the grade calculations are correct, even when the weights do not add up to 100%.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I can use AI to help me understand what errors I am making and explain the programming concepts I do not understand instead of having AI just give me the right answer straight away. This way I am still learning and not just copying and pasting code.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My goal is to make a working grade calculator that correctly calculates a student's current weighted grade and that i can use myself in the future. I also want to make a target grade feature.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to make a website to help college students calculate their grades. What features would help make it easier for students to calculate their grades?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

The main suggestion was to keep the project focused on calculating the students grades instead of a whole gradebook. The basic version should allow the user to enter grades and weights and calculate a weighted grade. An optional feature would be allowing the user to enter a target grade and calculate what grade they need on their remaining work. The project could use a simple interface with inputs and an output showing the current grade.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

The target-grade calculator might take me a while to code and I might not have the time to get done the main calculator and make a target-grade calculator, what is a simpler alternative?

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

I think keeping the project simple is better because it will give me the time I want to make sure the main calculator works correctly and i will add a edit and delete feature so you can change the assignment or grade if the user inputted the wrong value.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

If I get stuck I can use the course material, class examples, I can also ask the TA's and classmates for help. I can use AI to help explain an error or programming concept after trying to understand the problem myself and after going to my classmates and TA's.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

1.Read the error message.
2. Find the line of code where the error is occurring.
3. Check the variables and values being used on that line.
4. Test the function.
5. Use unit tests to check that the code is working properly.
6. If I still can not find the problem, i will ask a TA, a classmate, or AI for help to understand the error.
7. test the program again after making changes.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I want to make a website to help college students calculate their grades. What features would help make it easier for students to calculate their grades?

For a college grade-calculator website, I'd focus on features that answer the questions students actually have:

Useful features

1. **Enter grades**
   - Assignment/exam name
   - Points earned
   - Points possible
   - Example: `Midterm: 82 / 100`
2. **Current grade**
   - Automatically calculate the student's current percentage.
   - Show the corresponding letter grade, such as **87% → B+**.
3. **Weighted categories**
   - Allow students to enter categories like:
     - Homework — 20%
     - Quizzes — 20%
     - Exams — 60%
   - Then calculate the overall course grade.
4. **What-if calculator**
   - Let students enter a possible future grade and see how it would change their overall grade.
   - Example: *"If I get 90% on my next exam, what will my grade become?"*
5. **Target-grade calculator**
   - Student enters a desired final grade.
   - The website calculates what score they need on the remaining assignment/exam.
   - Example: *"What do I need on the final to finish with an A?"*

6. **Simple visual display**
    - A large current-grade percentage.
    - A progress bar or small chart could make the result easier to understand.
7. **Reset/edit assignments**
    - Students should be able to change or remove an assignment without starting completely over.

A simple version I'd recommend

For your project, I'd make the main flow:

Enter assignments → Calculate current grade → Try different future grades → See the new grade

The **what-if calculator** would be especially useful because it gives your website a purpose beyond simply doing arithmetic.

Since your project is being built with Drafter and is intended for an introductory programming course, I'd avoid things like student accounts, Canvas integration, databases, or automatically importing grades. Those would make the project much more complicated without necessarily making the basic calculator better.

The target-grade calculator might take me a while to code and I might not have the time to get done the main calculator and make a target-grade calculator, what is a simpler alternative?

Yes — I'd replace the target-grade calculator with something much simpler.

Best alternative: Grade summary

After the student enters their assignments, show a simple summary:

- **Current percentage:** 87.5%
- **Letter grade:** B+
- **Total points:** 175 / 200
- **Number of assignments:** 6

You could also add a **simple progress bar** showing where their current grade falls.

This is much easier than calculating what they need on a future assignment because you mainly need to:

1. Collect the assignment scores.
2. Add the points earned.
3. Add the points possible.
4. Divide earned points by possible points.
5. Convert the percentage into a letter grade.

6. Display the results.

One small extra feature

If you want your project to have something more interesting than just the basic calculation, I'd add **assignment editing/deleting**.

For example:

> Homework 1 — 18/20
> Quiz 1 — 8/10
> Midterm — 82/100
> **Current Grade: 82.3% (B-)**

The student could click **Edit** or **Delete** if they made a mistake.

That gives you a useful website without making the project unnecessarily complicated. It also fits nicely with your project assignment's focus on a clear **input** → **processing** → **output** structure.

If you're trying to finish within three weeks, I'd make your feature list:

Required: Add assignments + calculate grade + letter grade
Extra: Edit/delete assignments + simple visual grade display

That is a much safer scope than trying to build both a full calculator and a target-grade calculator.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.